

A Counterexample to Minimum Fano-Projection Selection for Five-Cycle Double Covers

Atharva Vaidya

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Scope and resolution status. This paper disproves a proposed *proof method*: selecting a cardinality-minimum binary projection of a nowhere-zero $\mathbb{F}_2^3$ -flow need not yield the component parity required by the Hušek–Šámal criterion. The constructed graph itself has an explicit five-cycle double cover. No counterexample to FiveCDC, and no resolution of FiveCDC, is claimed.

Abstract

Write a nowhere-zero $\mathbb{F}_2^3$ -flow on a cubic graph as $f = (h, s)$, where h is a binary flow and s is an $\mathbb{F}_2^2$ -flow. Call h an extendable projection. A recent flow criterion for the five-cycle double cover conjecture requires an extendable projection with a component-parity property, called clean here. A natural proposed selection principle was that some cardinality-minimum extendable projection should be cleanable.

We give a finite simple connected bridgeless cubic counterexample. The graph has 162 vertices and 243 edges. Its unique globally minimum extendable projection has 54 edges, is the disjoint union of two 27-circuits, and has no clean extension. The construction starts from an 18-vertex uncleanable projection and replaces eight support edges by an 18-vertex two-pole. Exact cycle-space enumeration proves that this two-pole has effective costs $w_0 = 7$ and $w_1 = 6$, with a unique minimum support in the 1-terminal state. A transparent weighted contraction argument then proves global minimality and uniqueness; uncleanability descends under contraction.

The negative result is strictly about the selection method. We give a human compositional five-cycle-double-cover construction for the same graph and a second literal assignment for the mandatory SAT/XOR encoding. Two independently written standard-library checkers reproduce the conditional minima, all 2^{14} lock placements, the uncleanability census, the canonical graph, and both positive cover certificates.

1 Conventions and the selection principle

All graphs in the theorem are finite, loopless, and cubic. Parallel edges would cause no difficulty in the flow statements, but the witness is simple. Edges are distinct objects. A *binary cycle* is an Eulerian edge set, possibly empty or disconnected. Thus “five Eulerian subgraphs” and “at most five, after appending empty members” are equivalent conventions.

Over a group of characteristic two, orientations may be suppressed: a flow $f : E(G) \rightarrow \mathbb{F}_2^k$ satisfies

$$\sum_{e \ni v} f(e) = 0 \quad (v \in V(G)).$$

It is nowhere zero if $f(e) \neq 0$ for every edge. Put $K = \mathbb{F}_2^2$ and write

$$f = (h, s) : E(G) \longrightarrow \mathbb{F}_2 \times K. \tag{1}$$

The support of the binary flow h is again denoted by h .

Definition 1.1. A binary cycle h is *extendable* if there is a K -flow s for which (h, s) is nowhere zero. Its minimum possible cardinality is

$$\mu(G) = \min\{|h| : h \text{ is an extendable projection}\}.$$

Fix an extension (h, s) . For $c \in K$, let

$$M_c = \{e \in h : s(e) = c\}. \quad (2)$$

At a cubic vertex no two incident edges can have the same nonzero full value, so every M_c is a matching. The four matchings partition h .

Let W be a component of $G - h$, and put

$$n_c(W) = |\delta(W) \cap M_c| \pmod{2}.$$

Cut conservation for the three coordinates implies that the four parities $n_c(W)$ are equal. This gives the following terminology.

Definition 1.2. An extension (h, s) is *clean* if $n_c(W) = 0$ for every component W of $G - h$, for one and hence every $c \in K$. The projection h is *cleanable* if it has a clean extension.

Hušek and Šámal prove that, for cubic graphs, existence of a clean extension is equivalent to existence of a five-cycle double cover [3, Theorem 3.16 and Conjecture 3.19]. Their formulation is related to Hoffmann-Ostenhof's necessary and sufficient condition for a prescribed 2-regular subgraph to occur in a five-cycle double cover [2]. The selection statement refuted here is

$$\text{Every bridgeless cubic graph has a cleanable cardinality-minimum extendable projection.} \quad (\text{S})$$

Statement (S) is not the FiveCDC conjecture itself.

2 The full-flow master lemma

The following elementary characterization makes “globally minimum” literal; in particular, it compares different low flows, not merely four affine rebases of one fixed extension.

Lemma 2.1 (full-flow master lemma). *Let G be a finite loopless cubic graph. For a K -flow s' , write $Z(s') = \{e : s'(e) = 0\}$. Then*

$$\mu(G) = \min_{\substack{s' \text{ a } K\text{-flow} \\ J \subseteq E - Z(s') \\ \partial J = \partial Z(s')}} (|Z(s')| + |J|), \quad (3)$$

where an infeasible inner problem has value $+\infty$. Every feasible zero set $Z(s')$ is a matching.

Proof. Let $M = Z(s')$, and suppose $J \subseteq E - M$ with $\partial J = \partial M$. Then $h' = M \dot{\cup} J$ has even degree at every vertex. The flow (h', s') is nowhere zero: the low value vanishes exactly on M , where the first coordinate is one.

Conversely, from any nowhere-zero (h', s') , its exact low zero set M is contained in h' . Put $J = h' - M$. The sets are disjoint and, since h' is binary, $\partial J = \partial M$. These constructions are inverse and preserve the objective $|h'| = |M| + |J|$, proving (3).

At a cubic vertex the zero-degree of a K -flow is 0, 1, or 3: two zero incident values force the third to be zero. Degree 3 is incompatible with a disjoint J having the same boundary. Hence a feasible M is a matching. $\square$

For fixed M , the inner problem is the ordinary shortest ∂M -join problem in $G - M$, with its standard odd-cut dual. The same dictionary isolates cleanliness.

Lemma 2.2 (the missing second join). *For a feasible triple (s', M, J_1) in Lemma 2.1, put $h = M \dot{\cup} J_1$. The extension (h, s') is clean if and only if there is a second ∂M -join*

$$J_2 \subseteq G - h. \quad (4)$$

Proof. A ∂M -join exists in a graph precisely when each component contains an even number of vertices of ∂M . For a component W of $G - h$,

$$|W \cap \partial M| \equiv |M \cap \delta(W)| = n_0(W) \pmod{2}.$$

The four cut-colour parities are equal by conservation, so their common vanishing is exactly the component condition for the join in (4). $\square$

Thus (S) asserts that some optimum branch of the disjunctive optimization (3) admits the extra packing (4). The graph below shows that this need not happen.

3 An uncleanable base projection

Let B be the 18-vertex cubic graph whose first fourteen edges are two disjoint 7-circuits:

$$(i, i + 1 \bmod 7) \ (0 \leq i < 7), \quad (7 + i, 7 + (i + 1 \bmod 7)) \ (0 \leq i < 7).$$

The remaining thirteen edges, in order, are

$$\begin{aligned} &(0, 11), (1, 9), (2, 12), (3, 10), \\ &(14, 4), (14, 5), (14, 15), (15, 6), (15, 16), \\ &(16, 13), (16, 17), (17, 7), (17, 8). \end{aligned} \quad (5)$$

Let $T = \{0, \dots, 13\}$, the union of the two 7-circuits. The low word

$$3231320|1320320|3112213212312 \quad (6)$$

extends T to a nowhere-zero $\mathbb{F}_2^3$ -flow.

Proposition 3.1 (base obstruction). *The projection T has 15,360 ordered low-coordinate extensions and none is clean.*

Exact finite proof. The cycle-space dimension of B is $27 - 18 + 1 = 10$. Enumerate its 1,024 binary cycles p, q . The pair extends T precisely when

$$E(B) - T \subseteq p \cup q. \quad (7)$$

There are 15,360 ordered pairs satisfying (7). The five components of $B - T$, restricted to vertices $0, \dots, 13$, have label word

$$01234444413024. \quad (8)$$

For every retained pair and every component, the checker xors the four cut-colour incidence rows. No pair makes all rows zero. This is a complete enumeration of all extensions by (7), not a sample or a solver heuristic. $\square$

The projection T is not globally minimum on B . We next make it uniquely minimum without changing the contraction obstruction.

4 A strict two-pole lock

Let S be the simple bridgeless cubic graph with graph6 encoding

$$\text{Q???C@?K?WEOM?_aGo?I__W@G?}. \quad (9)$$

Its ordered edges are

$$\begin{aligned} &(0, 7), (1, 8), (2, 9), (3, 9), (4, 10), (5, 10), (2, 11), (3, 11), (6, 11), \\ &(2, 12), (3, 12), (4, 12), (0, 13), (6, 13), (10, 13), (1, 14), (5, 14), \\ &(6, 14), (5, 15), (7, 15), (9, 15), (0, 16), (7, 16), (8, 16), \\ &(1, 17), (4, 17), (8, 17). \end{aligned} \quad (10)$$

Distinguish edge $e_* = e_5 = (5, 10)$. For $b \in \{0, 1\}$, let a_b be the minimum first-coordinate support size of a nowhere-zero $\mathbb{F}_2^3$ -flow on S , conditional on the first bit of e_* being b .

Lemma 4.1 (strict lock).

$$a_0 = 7, \quad a_1 = 5. \quad (11)$$

The a_1 -minimum support is unique and equals

$$\{5, 13, 14, 16, 17\}, \quad (12)$$

the 5-circuit $5 - 10 - 13 - 6 - 14 - 5$.

Exact finite proof. The cycle space of S again has dimension ten. A binary cycle h is extendable precisely when two cycles p, q satisfy $h \cup p \cup q = E(S)$. Enumerating the 1,024 choices of h and the 70,780 distinct exact zero sets generated by ordered (p, q) gives (11). There are nine a_0 -minimum supports, with 6,336 ordered extensions. In the $b = 1$ branch, (12) is the only minimum support and has 240 ordered extensions. A second checker constructs the cycle space by incidence-matrix elimination, rather than fundamental circuits, and obtains the same counts. $\square$

Delete e_* , expose its endpoints, and attach one new terminal link at each endpoint when the pole replaces an edge. If the common terminal first bit is 0, the least projected cost is

$$w_0 = a_0 = 7.$$

If it is 1, deletion removes the marked projected edge and the two links add two, so

$$w_1 = a_1 - 1 + 2 = 6. \quad (13)$$

This is the strict parity lock $w_0 > w_1$.

For completeness, the frozen full flow

$$(4, 6, 2, 6, 4, 5, 4, 2, 6, 6, 4, 2, 2, 3, 1, 2, 7, 5, 2, 6, 4, 6, 2, 4, 4, 6, 2) \quad (14)$$

attains (12), with the first coordinate represented by the least significant bit. Its marked value is $(1, 2) \in \mathbb{F}_2 \times K$. For any desired odd terminal value $(1, t)$, the invertible map

$$\Phi_t(x, \ell) = (x, \ell + x(2 + t)) \quad (15)$$

preserves the first coordinate and sends $(1, 2)$ to $(1, t)$. Thus the same unique projected pole support lifts every odd terminal value required by the base flow.

5 Eight locks and the 162-vertex theorem

For $L \subseteq T$, replace the base edges in L by disjoint fresh copies of the pole $S - e_*$; leave every other base edge unchanged. The target terminal support T has cost

$$14 + 5|L|. \quad (16)$$

For another extendable terminal projection H , put $O = T - H$ and $A = H - T$. The cost difference is

$$\Delta_L(H) = |O \cap L| - |O - L| + |A| = 2|O \cap L| - |O| + |A|. \quad (17)$$

Indeed, omitting a locked target edge costs one, omitting an unlocked target edge saves one, and adding a non-target base edge costs one.

Proposition 5.1 (lock-placement audit). *Among all $2^{14} = 16,384$ subsets L , precisely 2,280 make $\Delta_L(H) > 0$ for every extendable $H \neq T$. No such set has size at most seven. Exactly 180 have size eight. For*

$$L = \{0, 2, 3, 4, 7, 8, 9, 10\}, \quad (18)$$

the minimum strict gap is one.

Exact finite proof. The base cycle-space enumeration gives 1,008 extendable terminal supports. Formula (17) is evaluated for all 16,384 lock masks against the other 1,007 supports. The asserted counts and gap follow. An independent checker enumerates lock sets by combinations within each cardinality and reproduces the same 180 minimum masks. $\square$

Theorem 5.2. *There is a simple connected bridgeless cubic graph G on 162 vertices and 243 edges whose unique globally minimum extendable projection has size 54 and has no clean extension. The projection is the disjoint union of two 27-circuits.*

Proof. Use the eight substitutions in (18). Each pole contributes 18 fresh vertices and replaces one edge by 28, so

$$|V(G)| = 18 + 8 \cdot 18 = 162, \quad |E(G)| = 27 + 8 \cdot 27 = 243. \quad (19)$$

The construction is simple and cubic. The base and closure are bridgeless; equivalently, the literal checker deletes every edge of G in turn and retains connectivity.

Any nowhere-zero flow on G contracts through every pole. Summing conservation over the pole interior shows that its two nonzero terminal values agree. Restoring e_* gives a nowhere-zero flow on S , and contracting all poles gives one on B . Lemma 4.1 and (17) therefore lower-bound every projection.

The transformed flows (14)–(15) lift (6) and attain

$$14 + 5 \cdot 8 = 54.$$

Proposition 5.1 forces equality to have terminal support T . Equality in each pole then forces the unique local support (12). Hence the literal 54-edge projection is unique. Four locked edges lie on each original 7-circuit, and every replacement changes one support edge into a 6-edge path. Each therefore expands to length $7 + 4 \cdot 5 = 27$.

It remains to prove uncleanability. Deleting the 54-edge support leaves the five old components in (8) and eight pole-interior components. Suppose an extension upstairs were clean. Contract its poles. On every old component, a replaced support edge contributes exactly the colour on its endpoint link; this is the colour of the contracted edge. Thus all old component cut-colour parities are unchanged by contraction. A clean extension upstairs would therefore give a clean extension of T on B , contradicting Proposition 3.1. $\square$

6 The graph still has a five-cycle double cover

For every edge e and index $i \in \{0, \dots, 4\}$, introduce $x_{e,i} \in \{0, 1\}$. The standard mandatory encoding is

$$\sum_{i=0}^4 x_{e,i} = 2 \quad (e \in E), \quad \sum_{e \ni v} x_{e,i} = 0 \pmod{2} \quad (v \in V, 0 \leq i < 5). \quad (20)$$

The i -th selected edge set is Eulerian, and the edge equations give exact double coverage. Conversely every indexed five-cycle double cover gives such an assignment. At a cubic vertex the three incident two-subset labels are the three pairs on some three-element subset of $\{0, \dots, 4\}$.

There is a short human construction for G . In the edge orders (5) and (10), the following rows are five-cover pair labels:

$$\begin{aligned} B : & \quad 34 \ 23 \ 24 \ 14 \ 13 \ 12 \ 13 \ 03 \ 04 \ 02 \ 01 \ 04 \ 03 \ 02 \\ & \quad 14 \ 24 \ 34 \ 12 \ 34 \ 23 \ 24 \ 23 \ 34 \ 23 \ 24 \ 23 \ 34, \\ S : & \quad 24 \ 23 \ 12 \ 13 \ 13 \ 01 \ 14 \ 34 \ 13 \ 24 \ 14 \ 12 \ 04 \ 34 \\ & \quad 03 \ 34 \ 13 \ 14 \ 03 \ 02 \ 23 \ 02 \ 04 \ 24 \ 24 \ 23 \ 34. \end{aligned} \quad (21)$$

Each pair records the two cover indices containing that edge. Direct inspection at each cubic vertex checks that the three incident pairs form a triangle.

For a replaced base edge with pair P , permute the five indices on its copy of S so that the pair on e_* becomes P . Delete the base edge and e_* , and label both new terminal links by P . At every exposed endpoint, a removed incidence is replaced by the same pair, so all five parity equations remain even. Applying this operation independently to the eight locks proves:

Proposition 6.1. *The graph G of Theorem 5.2 has a five-cycle double cover.*

The artifact package also contains a separately generated literal assignment for all 243 edges. It is checked directly against (20) and against a 1,215-variable, 6,885-clause CNF expansion. The five class sizes in one independent certificate are

$$(92, 109, 97, 94, 94), \quad (22)$$

whose sum is $2|E(G)| = 486$. This is a positive SAT certificate, not an UNSAT claim.

7 Verification and trust boundary

The primary package freezes the labelled 243-edge list, canonical graph6 and sparse6 encodings, two five-cover certificates, the checker source, replay output, and a SHA-256 ledger. A second package rebuilds the graph in a different edge order. Its cycle-space code uses incidence-matrix row reduction instead of fundamental circuits, its bridge test uses Tarjan low links instead of repeated edge deletion, and its lock enumeration is organized by combinations instead of integer masks. The two constructed graphs canonicalize to the same graph6 record.

Quantity	Independently reproduced value
Closure cycles	1,024
Closure exact low-zero sets	70,780
(a_0, a_1)	(7, 5)
Base extendable supports	1,008
Base extensions of T	15,360
Clean base extensions	0
Lock masks	16,384
Minimum eight-lock masks	180
Order and size of G	162, 243
$\mu(G)$ and minimum-support ties	54, 1
Components of $G - H$	13
FiveCDC XOR rows	810

The human argument reduces the theorem to these finite counts and literal certificates. It does not replace exact enumeration of the two ten-dimensional cycle spaces or of the 2^{14} lock masks. The checkers use only the Python standard library; canonicalization uses Brendan McKay’s `labelg`. Full reproduction commands and frozen digests accompany the source.

8 Relation to prior work and novelty caution

The ordinary Cycle Double Cover Conjecture is distinct from the five-member strengthening. OpenAI announced a proof of the ordinary conjecture in July 2026 [6]; expositions by Oum and Geelen clarify that argument [4, 1]. Oum records the five-member version separately. The present graph has a five-member cover, so it conflicts with none of those results.

Hušek and Šámal give the immediate clean-flow criterion and state the five-member problem in that language [3]. Hoffmann-Ostenhof gives an earlier prescribed-subgraph characterization [2]. Jin, Mazzuoccolo, and Steffen study other Fano-flow conjectures through cores, joins, and weak oddness, and notably illustrate that natural strengthening principles in this area can fail [5]. Their objectives are not the minimum extendable binary projection $\mu(G)$ used here.

The master lemma is an elementary reparameterization of nowhere-zero $\mathbb{F}_2^3$ -flows by an exact zero set and a join; no priority is claimed for that identity. The specific strict-lock construction and the 162-vertex counterexample appear new to this project, but the literature screen was performed immediately after the Hušek–Šámal preprint appeared. It cannot establish priority or exclude unpublished parallel work. Expert comparison and independent human peer review are required before any novelty claim.

AI-use disclosure

OpenAI Codex agents, directed by Atharva Vaidya, developed the minimum-projection route, found and minimized the strict two-pole lock, implemented the enumerations, generated the certificates, and drafted this manuscript. Separate Codex agents wrote independent checkers and performed hostile audits. Atharva Vaidya directed the project and is responsible for deciding whether and how to disseminate the result. The work has not received independent human peer review. All finite claims, source code, literal graph encodings, and positive cover certificates are supplied for human verification.

References

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